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Original Article

Long-term outcomes after normothermic machine perfusion in liver transplantation—Experience at a single North American center

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ABSTRACT

Normothermic machine perfusion (NMP) has emerged as a valuable tool in the preservation of liver allografts before transplantation. Randomized trials have shown that replacing static cold storage (SCS) with NMP reduces allograft injury and improves graft utilization. The University of Alberta's liver transplant program was one of the early adopters of NMP in North America. Herein, we describe our 7-year experience applying NMP to extend preservation time in liver transplantation using a "back-to-base" approach. From 2015 to 2021, 79 livers were transplanted following NMP, compared with 386 after SCS only. NMP livers were preserved for a median time of minutes compared with minutes in the SCS cohort ($P < .0001$). Despite this, we observed significantly improved 30-day graft survival ($P = .030$), although there were no differences in long-term patient survival, major complications, or biliary or vascular complications. We also found that although SCS time was strongly associated with increased graft failure at 1 year in the SCS cohort ($P = .006$), there was no such association among NMP livers ($P = .171$). Our experience suggests that NMP can safely extend the total preservation time of liver allografts without increasing complications.

Abbreviations: AST, aspartate aminotransferase; BMI, body mass index; DBD, donation after brain death; DCD, donation after circulatory death; EAD, early allograft dysfunction; HA, hepatic artery; ICU, intensive care unit; INR, international normalized ratio; MELD-Na, model of end-stage liver disease sodium; NMP, normothermic machine perfusion; NRP, normothermic regional perfusion; POD, postoperative day; RCT, randomized control trial; SCS, static cold storage; UAH, University of Alberta Hospital.

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1. Introduction

Despite advances in surgical techniques, immunosuppressive therapy, and perioperative care over the past decades, a major limitation of solid organ transplantation remains the availability of suitable organs.[1] This has consequently led to the increased use of marginal grafts, such as those with prolonged cold ischemia times. In Canada, this has been particularly problematic, as the population is widely distributed (population density of 4 people/km^2) compared with Europe or the United States.[2] In an effort to enlarge the donor pool and improve graft quality for orthotopic liver transplantation, machine perfusion, as an alternative to the standard static cold storage (SCS), has been proposed as a means of extending preservation time without compromising graft function and potentially improving clinical outcomes.[3]

Several different types of machine perfusion (varying by temperature) have been explored for the preservation of liver grafts. Normothermic machine perfusion (NMP) preserves the organ at a physiologic temperature (~37 °C), which offers the opportunity for predictive metabolic assessment beyond preservation.[4] Evidence for its use has been bolstered by the large multicenter, randomized control trial (RCT) published by Nasralla et al[5] using the OrganOx metra device, which showed decreased organ injury (measured using aspartate transaminase [AST]) despite longer preservation time and an increased rate of graft utilization (ie, a lower rate of organ discard). Markmann et al[6] published similar findings in a subsequent RCT using the Organ Care Systems NMP device.

The University of Alberta Hospital was one of the first centers in North America to adopt this new technology in the setting of liver transplantation. Beginning in early 2015, our center began selectively utilizing NMP for the preservation of liver allografts before transplantation, which allows us to report the largest single-center experience in North America.[7] One limitation of our own approach has been an inability to transport this heavy and cumbersome technology between donor and recipient centers given the geography and our routine use of small planes to transport livers across vast distances, so we used NMP mostly in a “back-to-base” manner to avoid that challenge, except in local donors where the cold and transport time were short. Although convenient and most practical for us, this may have hampered the maximal benefit of this technology as there is an inherent obligate period of SCS before NMP. Herein, we present the clinical outcomes of patients transplanted with livers following NMP over 7 years compared with those transplanted after SCS only.

2. Materials and Methods

2.1. Study design

This study was designed as a phase I/II nonrandomized clinical trial to assess the safety of NMP in human livers prior to transplantation, as well as its efficacy in preventing organ injury. It was registered at clinicaltrials.gov with the study identifier NCT03089840. The study was approved by the Health Research Ethics Board at the University of Alberta and Health Canada (IRB 00043239, ID Pro00043239). All liver transplantations were carried out at the University of Alberta Hospital (UAH) in Edmonton, Alberta, Canada, from January 1, 2015, to December 31, 2021. The UAH is the primary liver transplant center for 2 Canadian provinces (Alberta and Saskatchewan), as well as northern British Columbia and the Northwest Territories, representing a catchment area of close to 6 million people. The follow-up was continued until June 30, 2022. Consent was obtained at the time of assessment for liver transplantation and confirmed on the date of transplant. Background characteristics collected included sex, age, body mass index (BMI), a model for end-stage liver disease sodium (MELD-Na) score, and reason for transplantation. The primary outcome was graft survival. Secondary outcomes included mortality, early allograft dysfunction (EAD; defined as having one or more of AST $>$ 2000 U/L on postoperative days [POD] 1 to 7, serum bilirubin $>$ 170 μmoL/L on POD7, or international normalized ratio $\geq$ 1.6 on POD7), intensive care unit (ICU) and hospital lengths of stay, and vascular and biliary complications. Only significant biliary and vascular complications requiring intervention (ie, $\geq$ Clavien-Dindo 3a) were included. Ischemic cholangiopathy was defined as the presence of nonanastomotic biliary strictures occurring with a patent, nonstenotic hepatic artery. Outcomes were compared with those of liver transplant recipients whose transplants occurred during the same period but whose graft was preserved using SCS only. Living donor, pediatric, and domino liver transplantations were excluded from the SCS cohort. Five livers subjected to NMP that were subsequently discarded were excluded from the analysis (Supplementary Table 1).

2.2. Normothermic machine perfusion

Livers allocated for NMP at the discretion of the operating surgeon (eg, for considerations of bed or patient availability, to avoid operating at odd hours or when 2 livers were available at the same time) were preserved using the OrganOx metra system. Livers were procured in the standard fashion, including *in situ* flushing with cold histidine-tryptophan-ketoglutarate (Custodiol) or the University of Wisconsin solution (Bridge to Life). Upon procurement, organs were stored in histidine-tryptophan-ketoglutarate or the University of Wisconsin solution at 4 °C for transportation from the donor center to the UAH. NMP was initiated promptly with minimal cold storage for a subset of local donors. Once received at the UAH, the OrganOx was primed with 500 mL of gelofusine (B. Braun Canada Ltd) and 3 units of type-O packed red blood Co. Boluses of antibiotics (cefuroxime 750 mg; GlaxoSmithKline Canada), heparin (10,000 U; Fresenius Kabi Canada), and calcium gluconate (10%, 10 mL; Fresenius Kabi Canada).[8] The liver was then flushed with 1 L of gelofusine and 500 mL of 5% human albumin prior to attaching it to the circuit.

During NMP, the livers were monitored with point-of-care blood gas analysis, as well as biochemical analysis of perfusate (eg, alanine aminotransferase, AST, and bilirubin) performed by the clinical laboratory at the onset of NMP and every 2 hours thereafter. The livers received infusions of heparin (833 U/h; Fresenius Kabi Canada), epoprostanol (8 μg/h;

GlaxoSmithKline Canada), insulin (6.7 U/h; Eli Lilly Canada Inc), sodium taurocholate (0.19 g/h; New Zealand Pharmaceuticals Ltd), and nutrition (Nutriflex, B. Braun Canada Ltd).[8] Glucose was maintained within physiological parameters (5-8 mmoL/L) by adjusting the Nutriflex infusion, and sodium bicarbonate (8.4%; Hospira Inc) was used to maintain a physiological pH (7.35-7.45). Once the recipient hepatectomy was performed, the livers were disconnected from perfusion and flushed with a cold Ringer's lactate solution immediately before implantation in compliance with clinical practice.

2.3. Perioperative management

Implantation of the liver graft occurred in a standard fashion in both groups. Livers were transplanted with a standard caval replacement without bypass or a temporary portocaval shunt. Patients were transferred to the ICU immediately postoperatively and subsequently extubated and transferred to a standard ward bed depending on their clinical condition. Standard protocols for tacrolimus-based immunosuppression were initiated post-operatively, with some patients receiving sirolimus due to pre-existing renal dysfunction.

2.3. Statistical analysis

Statistical analysis was performed using Stata 16.1 (Stata-Corp LLC). Data are presented as medians with interquartile ranges or proportions, where appropriate. Comparisons between medians were performed using Mann-Whitney U tests, whereas comparisons between proportions used the chi-squared test. To eliminate the effect of confounding on the relationship between storage type and several outcome variables, transplanted livers from the SCS cohort were matched 1:1 with livers from the NMP cohort based on MELD-Na, donor risk index, donor and recipient age, and donor type (ie, donation after circulatory death [DCD] vs brain death [DBD]).[9] Additionally, we employed multivariate logistic regressions using a hypothesis-driven purposeful selection methodology, in which bivariate analysis of variables with a P value $< .1$ or from variables previously deemed clinically relevant to our primary outcome was used to generate a main effects model. Kaplan-Meier survival curves were plotted for graft and patient survival, as well as biliary and vascular complication-free survival, using the log-rank test for comparison. The same statistical methods were used to compare matched cohorts and subsets of cohorts.

3. Results

During the study period, 80 orthotopic liver transplantations were performed following NMP in 79 recipients. A single patient withdrew consent after being transplanted and was excluded from the study. Over the same period at our center, 386 liver transplants were performed on 370 recipients, excluding pediatric, living donor, and domino graft transplantation. The median follow-up time from the first transplant was 3.77 (1.63-4.52; max 7.17) years for patients in the NMP cohort and 3.00 (1.26-5.19; max 7.43) years for patients in the SCS cohort ($P = .965$).

3.1. Baseline donor and recipient demographics

Most donor characteristics were similar between groups, including age, sex, and BMI (Table 1). However, the groups differed slightly in the proportion of donors who had died from an overdose (19.4% of the SCS cohort vs 10.1% of the NMP cohort; $P = .049$), as well as other causes (eg, primary central nervous system malignancy, medical assistance in dying), which made up 5.2% of donors in the SCS cohort compared with 13.9% in the NMP cohort ($P = .005$). The NMP group had a higher proportion of DCD donors, although this was not statistically significant (13.2% in the SCS cohort vs 20.3% in the NMP cohort; $P = .104$). Median SCS time was significantly longer in the NMP group (288.5 [193-383] minutes in the SCS cohort vs 359 [301-412] minutes in the NMP cohort; $P = .0001$). The median warm perfusion time was 475 (387-620) minutes, leading to a total preservation time of 847 (767-964) minutes in the NMP cohort (compared with 289 [193-383] minutes in the SCS group; $P < .0001$) (Fig. 1). Recipients were similar in their characteristics, with no difference in age, sex, BMI, the reason for transplant, or the proportion of retransplant patients between the groups. The NMP cohort had a slightly higher proportion of multiorgan transplant recipients (kidney or heart), although this was not statistically significant owing to the small numbers in both groups (1.0% in the SCS cohort vs 3.8% in the NMP cohort; $P = .067$).

No significant differences were observed among donors or recipients following DCD donation, with the exception of total preservation time, which was higher in the NMP cohort (317 [259-383] minutes in the SCS cohort vs 965 [734.5-1217.5] minutes in the NMP cohort; $P < .0001$) (Supplementary Table 2). Similarly, local donors and their recipients were not different between the cohorts. Both total and cold preservation times were significantly longer for livers in the NMP cohort (223 [140-292] minutes and 818 [689-911] minutes, respectively) compared with those in the SCS cohort (295 [99-209] minutes; $P = .003$ and $P < .0001$, respectively) (Supplementary Table 3).

3.2. Unadjusted clinical outcomes

Primary and secondary clinical outcomes are presented in Table 2. Graft survival after 30 days was significantly improved in the NMP cohort (94.3% in the SCS cohort vs 100% in the NMP cohort; $P = .030$); however, there was no difference at later time points (6 months: 90.1% in the SCS cohort vs 89.9% in the NMP cohort, $P = .945$; 1 year: 89.1% in the SCS cohort vs 86.8% in the NMP cohort, $P = .571$; 3 years: 77.5% in the SCS cohort vs 79.3% in the NMP cohort, $P = .766$). Overall, graft survival was not significantly different between cohorts ($P = .845$), as seen in the survival curve (Fig. 2A and Supplementary Table 4). Recipient survival was also not different between cohorts ($P = .936$; Fig. 2B and Supplementary Table 4). Secondary clinical outcomes, including early allograft dysfunction (28.2% in the SCS cohort vs 25.3% in the NMP cohort; $P = .597$) were not significantly different. Median ICU and total hospital lengths of stay were also comparable (3 [2-8] days in the SCS cohort vs 4 [2-11] days in the NMP cohort, $P = .129$, and 17 [12 -32] days in the SCS cohort vs 19 [12-37]

Table 1
Donor and recipient demographics.

Background characteristics	SCS (n = 386)	NMP (n = 79)	*P* value
Donor characteristics			
Age, y	37 (26-53)	40 (25-56)	.526
Female (%)	144 (37.3)	27 (34.2)	.599
BMI	25.8 (22.3-29.4)	25.7 (22.4-29.2)	.765
Cause of death (%)			
Intracranial event	109 (28.2)	22 (27.9)	.944
Anoxia/hypoxia	104 (26.9)	21 (26.6)	.947
Trauma	78 (20.2)	17 (21.5)	.792
Overdose	75 (19.4)	8 (10.1)	.049
Other	20 (5.2)	11 (13.9)	.005
Local (%)	139 (36.0)	23 (29.1)	.241
DCD (%)	51 (13.2)	16 (20.3)	.104
Preservation details			
Functional warm ischemia time, min[a]	19 (12-27)	21.5 (15-25.5)	.527
Cold ischemia time, min	288.5 (193-383)	359 (301-412)	.0001
NMP time, min	N/A	475 (387-620)	N/A
Rewarm time, min	40 (35-46)	40 (36-47)	.280
Total storage time, min	288.5 (193-383)	847 (767-964)	< .0001
Recipient characteristics			
Age, y	56.0 (45.7-62.4)	58.1 (46.9-63.1)	.845
Female (%)	118 (30.6)	23 (29.1)	.798
BMI	27.5 (24.2-32.1)	28.0 (23.8-35.0)	.578
MELD-Na (at the time of transplant)	19 (13-27)	16 (11-23)	.064
Creatinine, μmoL/L	84 (65-111)	84 (65-103)	.843
Diagnosis (%)			
EtOH cirrhosis	125 (32.4)	25 (31.7)	.898
Hep C cirrhosis	93 (24.1)	20 (25.3)	.817
Hep B cirrhosis	17 (4.4)	6 (7.6)	.233
NASH cirrhosis	62 (16.1)	12 (15.2)	.847
HCC	132 (34.2)	25 (31.7)	.662
Autoimmune	23 (6.0)	4 (5.1)	.757
PBC	26 (6.7)	4 (5.1)	.581
PSC	37 (9.6)	8 (10.1)	.882
Retransplant (%)	34 (8.8)	7 (8.9)	.988
Multiorgan transplant (%)	4 (1.0)	3 (3.8)	.066

BMI, body mass index; DCD, donation after circulatory death; EtOH, ethanol; HCC, hepatocellular carcinoma; Hep B, hepatitis B; Hep C, hepatitis C; MELD, model of end-stage liver disease; N/A, not applicable; NASH, nonalcoholic steatohepatitis; NMP, normothermic machine perfusion; PBC, primary biliary cholangitis; PSC, primary sclerosing cholangitis; SCS, static cold storage.

[a] Applies to DCD livers only; values presented with an interquartile range where applicable.

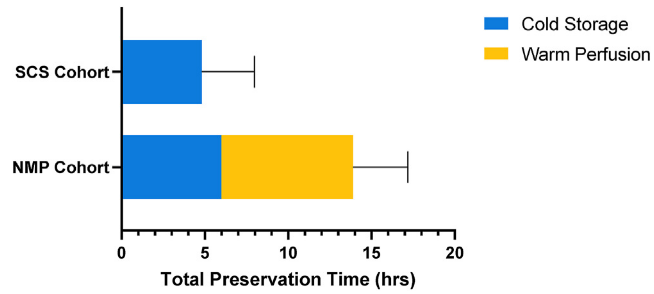

Figure 1. Median total storage times for liver grafts in the static cold storage only (SCS) and normothermic machine perfusion (NMP) cohorts ($P < .0001$).

days in the NMP cohort, $P = .258$, respectively). Major complications (Clavien-Dindo classification ≥ 3b) were likewise no different between cohorts (47.9% in the SCS cohort vs 54.4% in the NMP cohort; $P = .292$). Biliary and vascular complications requiring intervention (Clavien-Dindo classification ≥ 3a) within 1 year were also no different between cohorts (15.4% in the SCS cohort vs 21.5% in the NMP cohort, $P = .193$, and 11.9% in the SCS cohort vs 16.5% in the NMP cohort, $P = .269$, respectively). Similarly, the timing of biliary and vascular complications requiring intervention was not different ($P = .225$ and .366, respectively, Fig. 3A, B). Specifically, there was no difference in ischemic cholangiopathy between the cohorts (1.0% in the SCS cohort vs 1.3% in the NMP cohort; $P = .857$). No patients in the NMP cohort experienced primary graft nonfunction compared to 9 patients in the SCS cohort; however, this was not significant ($P = .170$) owing to the small numbers overall.

Following the DCD donation, the majority of outcomes were not different between the cohorts (Supplementary Table 5). Peak AST within the first 7 postoperative days was significantly lower in the NMP cohort among DCD recipients (2600 [1279-2600] U/L in the SCS cohort vs 1700 [606-2284.5] U/L in the NMP cohort; $P = .045$), which is reflected in the decreased rate of EAD (though this did not achieve significance; 63.7% in the SCS cohort vs 43.8% in the NMP cohort, $P = .136$). Although 30-day graft survival was again higher in the SCS cohort, including only DCD donors, this was not statistically significant owing to smaller numbers (92.2% in the SCS cohort vs 100% in the NMP cohort; $P = .248$). Clinical outcomes were similar for local donors between the 2 cohorts (Supplementary Table 6).

Table 2
Comparison of clinical outcomes between liver transplantation after static cold storage vs normothermic machine perfusion.

Recipient outcomes	SCS (n = 386)	NMP (n = 79)	*P* value
Crude graft survival			
30 d (n = 386/79)	364 (94.3)	79 (100.0)	.030
6 mo (n = 385/79)	347 (90.1)	71 (89.9)	.945
1 y (n = 349/76)	311 (89.1)	66 (86.8)	.571
3 y (n = 249/58)	193 (77.5)	46 (79.3)	.766
5 y (n = 168/25)	104 (61.9)	11 (44.0)	.089
7 y (n = 16/3)	12 (75.0)	2 (66.7)	.764
Crude patient survival[a]			
30 d (n = 369/76)	358 (97.0)	76 (100.0)	.127
6 mo (n = 368/76)	344 (93.5)	71 (93.4)	.985
1 y (n = 331/73)	307 (92.8)	67 (91.8)	.775
3 y (n = 237/55)	193 (81.4)	46 (83.6)	.703
5 y (n = 155/22)	104 (67.1)	11 (50.0)	.116
7 y (n = 16/3)	12 (75.0)	2 (66.7)	.764
EAD (%)	109 (28.2)	20 (25.3)	.597
Peak AST POD1-7, U/L	695.5 (296.5-1519.5)	813 (358-1610)	.323
Total bilirubin POD7, µmol/L	37 (21-86)	42 (21-94)	.705
INR POD7	1.1 (1.0-1.2)	1.1 (1.0-1.2)	.019
Primary nonfunction	9 (2.3)	0 (0.0)	.171
ICU length of stay (d)	3 (2-8)	4 (2-11)	.129
Hospital length of stay (d)	17 (12-32)	19 (12-37)	.258

(*continued on next page*)

Table 2 (*continued*)

Recipient outcomes	SCS (n = 386)	NMP (n = 79)	*P* value
Major complications (Clavien-Dindo ≥3b)	185 (47.9)	43 (54.4)	.292
Biliary complications within 1 y	60 (15.4)	17 (21.5)	.193
Anastomotic stricture	39 (10.1)	12 (15.2)	.187
Leak	20 (5.2)	4 (5.1)	.966
Ischemic cholangiopathy	4 (1.0)	1 (1.3)	.857
Biliary complications >1 y (n = 312/66)	6 (1.9)	3 (4.6)	.204
Vascular complications at 1 y	46 (11.9)	13 (16.5)	.269
HA stenosis	19 (4.9)	7 (10.1)	.072
HA thrombosis	7 (1.8)	1 (0.9)	.733
Rejection episode within 1 y	74 (19.2)	16 (20.3)	.824

AST, aspartate transaminase; EAD, early allograft dysfunction; HA, hepatic artery; ICU, intensive care unit; INR, international normalized ratio; POD, postoperative day; NMP, normothermic machine perfusion; SCS, static cold storage.

[a] Excludes patients retransplanted during the study period; values presented with interquartile range where applicable.

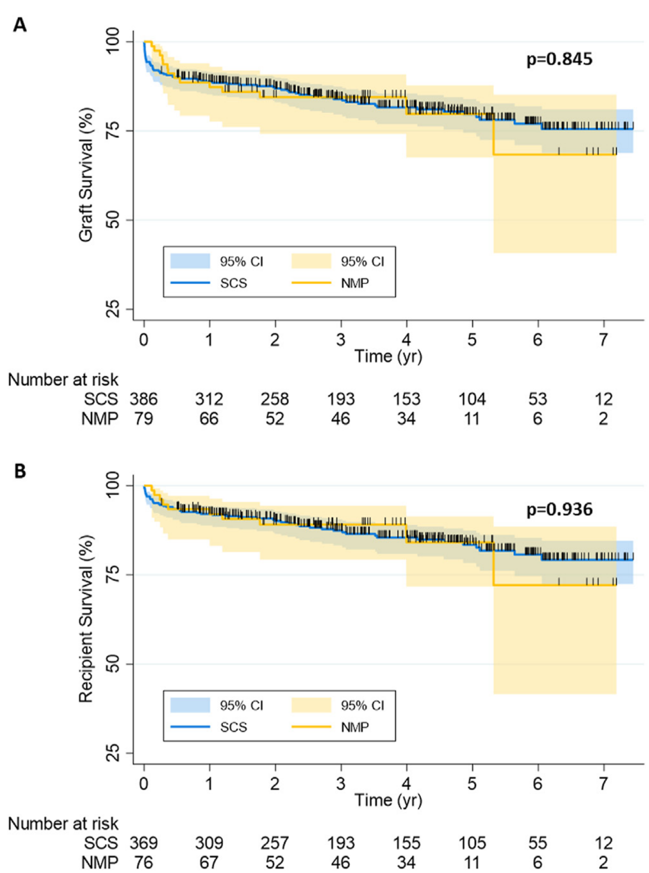

Figure 2. Kaplan-Meier curve showing graft (A) and the recipient (B) survival after either static cold storage (SCS) or normothermic machine perfusion (NMP) preservation.

3.3. Matched clinical outcomes

Several key clinical parameters were used to match transplants in the SCS cohort with those in the NMP cohort. Matched cohorts were no different in terms of baseline characteristics of donors and recipients, although SCS time and total preservation time remained significantly different (P = .013 and P < .0001, respectively, Supplementary Table 7). Thirty-day graft survival was similarly significantly improved in the NMP cohort after matching (92.4% in the SCS cohort vs 100% in the NMP cohort; P = .013) (Table 3). Thirty-day recipient survival was also significantly improved comparing matched cohorts (94.7% in the SCS cohort vs 100% in the NMP cohort; P = .041). However, other clinical and long-term outcomes remained unchanged. Control for confounding using multivariate logistic regression modeling similarly did not result in significant differences in clinical outcomes between cohorts (Supplementary Tables 8 and 9).

3.4. Impact of cold ischemia time

Cold ischemia time ranged from 53 to 920 minutes in the SCS group and 66 to 688 minutes in the NMP group. Among recipients in the SCS cohort, those with graft failure at 30 days, 6 months, and 1 year had significantly longer median SCS times compared with those with graft survival at the same time points (30 days: 286 [190-382] minutes for those without graft failure vs 387 [295-472] minutes for those with, P = .003; 6 months: 288 [189-383] minutes for those with graft failure vs 354 [266-527] minutes for those with, P = .006; 1 year: 286 [187-381.5] minutes for those without graft failure vs 354 [281-481] minutes for those with, P = .005). However, this association was not observed in the NMP cohort, among whom SCS times were not significantly longer for those with graft failure at 6 months or 1 year compared with those with graft survival (6 months: 359 [292-410] minutes for those without graft failure vs 422 [350-523] minutes for those with, P = .064; 1 year: 359 [292-412] minutes for those without graft failure vs 378 [332-480] minutes for those with, P = .171). Neither was NMP time associated with graft failure at 6 months or 1 year in the NMP cohort (P = .889 and P = .631, respectively). Both graft and recipient survival decreased in the SCS cohort for livers with the highest quartile of SCS time (≥397 minutes)

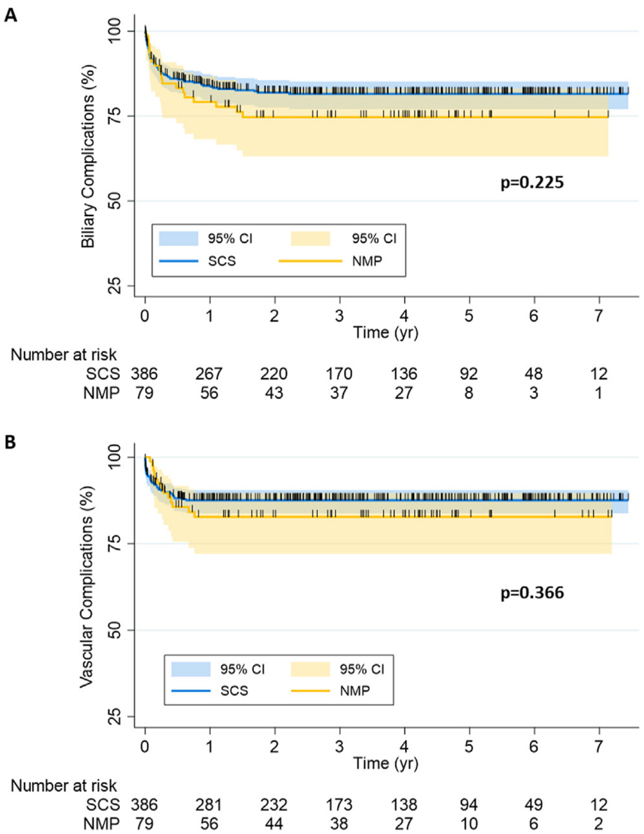

Figure 3. Kaplan-Meier curve showing time to biliary (A) and vascular (B) complications requiring intervention.

compared with those with lower SCS times ($P = .007$ and .049, respectively, Fig. 4A, B). However, this was not observed in the NMP cohort ($P = .362$ and .606, respectively, Fig. 4C, D).

Recipients with EAD also had longer median SCS times in the SCS cohort (281 [166-370] minutes for those without EAD vs 320 [250-416] minutes for those with; $P = .002$), but not in the NMP cohort (364 [323-415] minutes for those without EAD vs 350 [296.5-390] minutes for those with; $P = .278$). Correspondingly, the proportion of EAD was significantly increased for recipients of livers with SCS time in the highest quartile in the SCS cohort (39.1% in the 4th quartile vs 24.8% in quartiles 1 to 3; $P = .009$), whereas, in the NMP cohort, this was almost reversed (14.8% in the 4th quartile vs 30.8% in quartiles 1 to 3; $P = .122$). There was also a moderate association between SCS time and peak AST in the first postoperative week ($\rho = 0.284$; $P < .0001$) in the SCS cohort, but no association in the NMP cohort ($\rho = -0.026$; $P = .820$). In contrast, the length of NMP storage time, which ranged from 196 to 1347 minutes, was not associated with EAD (475 [390-618] minutes for those without EAD vs 481 [345-662] minutes for those with; $P = .847$) or peak AST ($\rho = -0.019$; $P = .867$).

Looking at a subset of grafts in the SCS cohort with prolonged SCS times (≥8 hours) compared to NMP grafts with similar total preservation times (ie, ≥8 hours), the impact of cold ischemia becomes apparent. Graft survival was significantly higher at 30 days for NMP livers compared to SCS livers preserved for ≥8 hours (88.6% in the SCS cohort vs 100% in the NMP cohort; $P = .003$), with a trend toward higher graft survival at 6 months (77.3% in the SCS cohort vs 89.6% in the NMP cohort; $P = .067$)

Table 3

Comparison of clinical outcomes between liver transplantation after static cold storage vs normothermic machine perfusion matched 1:1.

Matched recipient outcomes	SCS (n = 79)	NMP (n = 79)	*P* value
Crude graft survival			
30 d (n = 79/79)	73 (92.4)	79 (100.0)	.013
6 mo (n = 78/79)	69 (88.5)	71 (89.9)	.776
1 y (n = 71/76)	62 (87.3)	66 (86.8)	.931
3 y (n = 48/58)	35 (72.9)	46 (79.3)	.440
5 y (n = 33/25)	18 (54.6)	11 (44.0)	.426
Crude patient survival[a]			
30 d (n = 75/76)	71 (94.7)	76 (100.0)	.041
6 mo (n = 74/76)	67 (90.5)	71 (93.4)	.516
1 y (n = 66/73)	60 (90.9)	67 (91.8)	.855
3 y (n = 47/55)	35 (74.5)	46 (83.6)	.254
5 y (n = 32/22)	18 (56.3)	11 (50.0)	.651
EAD (%)	26 (32.9)	20 (25.3)	.293
Peak AST POD1-7, U/L	697 (293-1810)	813 (358-1610)	.894
Total bilirubin POD7, µmol/L	43.5 (26.5-98)	42 (21-94)	.590

(*continued on next page*)

Table 3 (*continued*)

Matched recipient outcomes	SCS (n = 79)	NMP (n = 79)	*P* value
INR POD7	1.1 (1.0-1.2)	1.1 (1.0-1.2)	.182
Primary nonfunction	2 (2.5)	0 (0.0)	.155
ICU length of stay (d)	3 (2-8)	4 (2-11)	.163
Hospital length of stay (d)	17 (12-34)	19 (12-37)	.236
Major complications (Clavien-Dindo ≥3b)	43 (54.4)	43 (54.4)	1.000
Biliary complications within 1 y	12 (15.2)	17 (21.5)	.304
Anastomotic stricture	8 (10.1)	12 (15.2)	.339
Leak	5 (6.3)	4 (5.1)	.731
Ischemic cholangiopathy	1 (1.3)	1 (1.3)	1.000
Biliary complications >1 y	1 (1.3)	3 (3.8)	.311
Vascular complications within 1 y	10 (12.7)	13 (16.5)	.499
HA stenosis	3 (3.8)	7 (10.1)	.118
HA thrombosis	1 (1.3)	1 (1.3)	1.000
Rejection episode within 1 y	14 (17.7)	16 (20.3)	.685

AST, aspartate transaminase; EAD, early allograft dysfunction; HA, hepatic artery; ICU, intensive care unit; INR, international normalized ratio; POD, postoperative day; NMP, normothermic machine perfusion; SCS, static cold storage.

[a] Excludes patients retransplanted during the study period; values presented with an interquartile range where applicable.

and 1 year (72.1% in the SCS cohort vs 86.5% in the NMP cohort; *P* = .055; Table 4). Similarly, recipient survival was significantly higher at 30 days (92.7% in the SCS cohort vs 100% in the NMP cohort; *P* =.018) and 1 year (77.5% in the SCS cohort vs 91.6% in the NMP cohort; *P* = .038), with a trend toward improved survival at 6 months (82.9% in the SCS cohort vs 93.2% in the NMP cohort; *P* = .083). Both of these trends for improved graft and recipient survival are apparent using NMP for extended preservation compared with SCS over the duration of the study (Fig. 5).

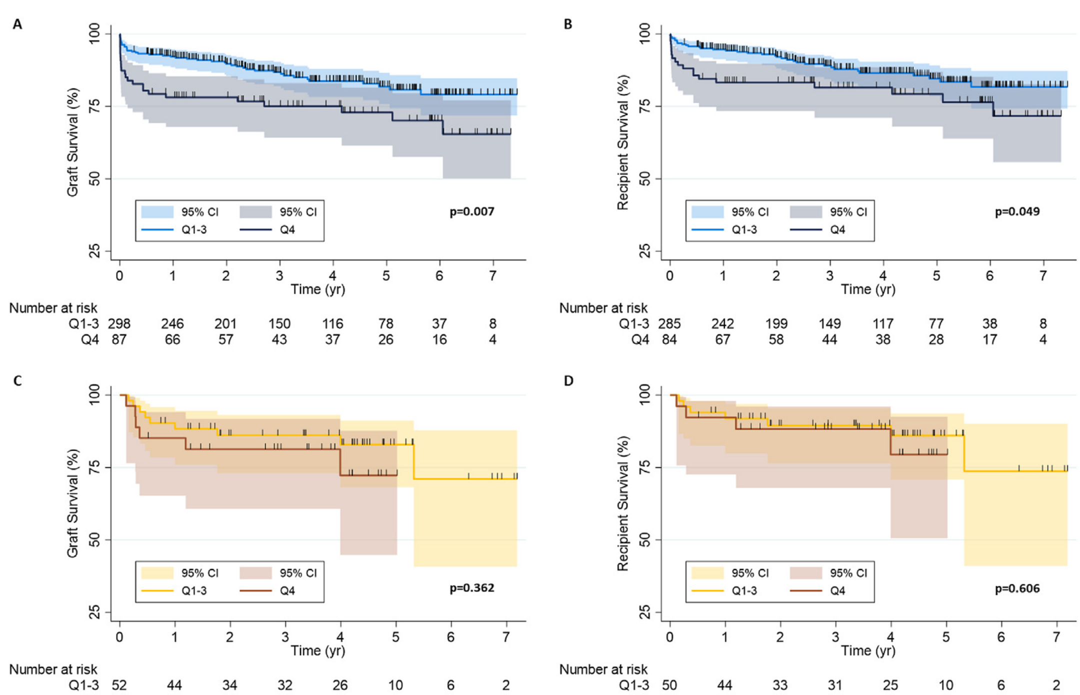

Figure 4. Graft (A) and the recipient (B) survival between livers with the highest quartile of cold preservation time in the static cold storage cohort. Graft (C) and the recipient (D) survival between livers with the highest quartile of cold preservation time in the normothermic machine perfusion cohort.

Table 4
Comparison of clinical outcomes between liver transplantation after static cold storage vs normothermic machine perfusion for transplanted livers with a total preservation time of 8 h or greater.

Recipient outcomes	SCS (n = 44)	NMP (n = 77)	*P* value
Crude graft survival			
30 d (n = 44/77)	39 (88.6)	77 (100.0)	.003
6 mo (n = 44/77)	34 (77.3)	69 (89.6)	.067
1 y (n = 43/74)	12 (72.1)	66 (86.5)	.055
3 y (n = 34/56)	22 (64.7)	44 (78.6)	.440
Crude patient survival[a]			
30 d (n = 41/74)	38 (92.7)	76 (100.0)	.018
6 mo (n = 41/74)	34 (82.9)	69 (93.2)	.083
1 y (n = 40/71)	31 (77.5)	65 (91.6)	.038
3 y (n = 32/53)	22 (68.8)	44 (83.0)	.126
EAD (%)	16 (36.4)	19 (24.7)	.173
Peak AST POD1-7, U/L	1297 (587-2252)	760 (358-1606)	.043
Total bilirubin POD7, μmol/L	41 (30.5-100)	42 (21-94)	.547
INR POD7	1.1 (1.0-1.1)	1.1 (1.0-1.2)	.026
Primary nonfunction	3 (6.8)	0 (0.0)	.020

AST, aspartate transaminase; EAD, early allograft dysfunction; INR, international normalized ratio; POD, postoperative day; NMP, normothermic machine perfusion; SCS, static cold storage.

[a] Excludes patients retransplanted during the study period; values presented with an interquartile range where applicable.

4. Discussion

Our study represents one of the earliest and longest-duration single-center experiences with liver transplantation after NMP. We found that the addition of NMP following traditional SCS had no adverse effect on multiple clinical outcomes, including mortality and biliary and vascular complications, and showed a clear benefit in terms of early allograft survival. While extended preservation time under SCS conditions results in higher rates of graft failure and mortality, our study showed that extending preservation time with the protection of NMP does not impede clinical outcomes.

The efficacy of NMP over SCS has been demonstrated convincingly by 2 multicenter RCTs using different NMP devices with 1-year follow-up (ie, OrganOx metra and Organ Care Systems liver).[5,6] In these studies, livers were recovered in the traditional fashion and placed on the NMP circuit as soon as possible. This resulted in the NMP groups having significantly lower SCS time, despite having longer total preservation times. In contrast, our study had significantly longer SCS times in the NMP cohort, in addition to having overall longer preservation times, reflecting our "back-to-base" more practical approach. This may explain why we did not find significantly lower rates of EAD in the NMP cohort, although an additional subanalysis of local donors (Supplementary Table 6) with shorter SCS times did not alter longer-term outcomes in our series. In addition, our study also included retransplants (ie, patients receiving second or third liver transplantations) and multiorgan transplants, which are more prone to complications.[10] Interestingly, we did find improved 30-day graft survival and no instances of PNF in the NMP cohort, despite longer cold and total preservation times. The fact that graft survival was no different at later time points emphasizes that the impact of NMP is greatest early on. Over time, these initial effects are likely to be diluted by other circumstances arising in the posttransplant period.

The results from our center reflect a practical way of applying NMP in a North American geographically dispersed environment, avoiding the additional logistical challenges of transporting heavy and complex equipment by plane and ground ambulance at each end. We demonstrate how NMP can be used to safely extend the preservation of a liver allograft to avoid operating at late hours or accommodating delays due to other surgeries or ICU bed availability (which was problematic during the coronavirus disease 2019 pandemic). This strategy is also useful with challenging recipients, such as more complex reoperative, retransplantation, or multiorgan transplantation, where explantation of the diseased organ(s) may be prolonged.[11] Unlike extended preservation under SCS conditions, extending preservation with NMP does not increase the risk of graft failure or recipient mortality, which is evident in our SCS cohort. However, the application of NMP does not positively improve long-term outcomes in our experience.

Although clinically approved NMP devices are designed to be portable, they are still more cumbersome than using cold storage and require trained personnel to operate. The OrganOx metra, in

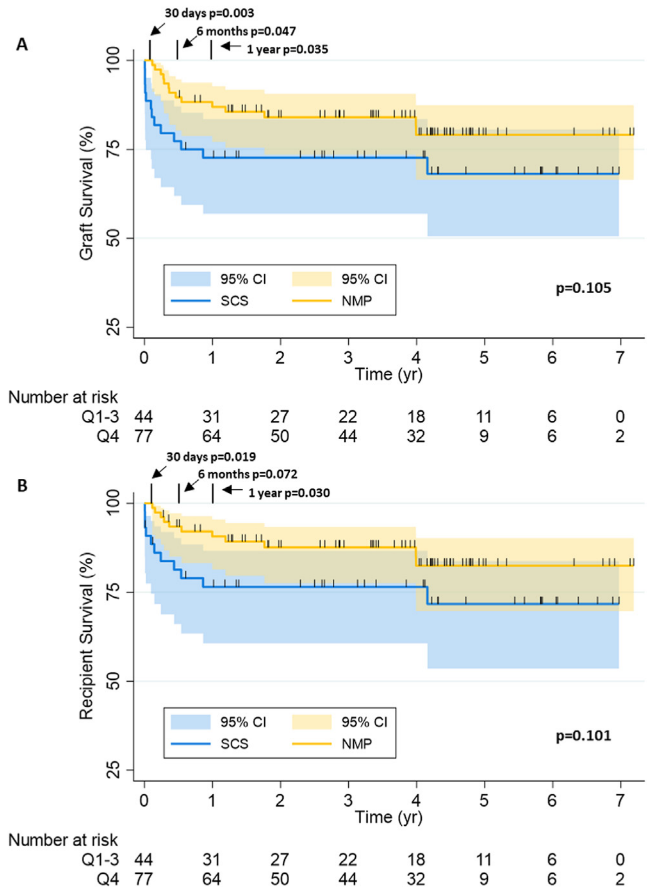

Figure 5. Graft (A) and recipient (B) survival curves for livers preserved for ≥8 h prior to transplantation, compared between static cold storage (SCS) and normothermic machine perfusion (NMP) cohorts.

particular, is designed to be portable in the back of small van-type ambulances used in the United Kingdom and Europe. Because of our unique geographic circumstances (being situated nearly 1200 km from the next closest liver transplant center), a high percentage of organs arrived from distant sites by air, and consequently, we have adopted a "back-to-base" approach for integrating NMP into our liver procurement strategy.[12] Though shorter SCS times are associated with improved outcomes, it is probably not necessary or cost effective to transport every liver under NMP, and it has not been clearly established which livers would benefit most from NMP. The observation that extended SCS was associated with decreased graft and patient survival in the SCS cohort but not in the NMP is a unique finding from our study that suggests that NMP may be particularly beneficial in these organs. This hypothesis merits consideration in future clinical trials. Normothermic regional perfusion (NRP), in which abdominal organs are perfused within the body following controlled DCD, may offer an alternative or complementary approach with DCD livers, as it avoids the challenges of transporting the organ while it is being perfused.[13] Retrospective studies from several centers in Europe have reported decreased biliary complications and reduced graft loss with this approach.[14,15]

An additional consideration with NMP is the impact of the artificial *ex vivo* environment on the organ, which, although an improvement over SCS conditions, can still produce proinflammatory responses.[16] These effects may be more pronounced on injured livers, such as those donated after circulatory death. Current commercial devices also cannot fully replicate the *in vivo* environment, lacking, for instance, mechanisms to eliminate metabolic waste (ie, as a replacement for the kidney), as well as hormonal and neurologic input. Still, it seems that the liver, particularly a healthy one, can be maintained reliably for ≤24 hours with commercially available devices, with emerging modifications continuing to push the limits of preservation.[17]

The limitations of this study include its nonrandomized nature. Although our use of 1:1 matching to account for some potential confounders did not change our conclusions, it is possible that residual confounding exists that may have affected outcomes. Given the retrospective nature of our SCS cohort and that biliary lesions are not routinely assessed at our center in the absence of symptoms, we were unable to comment on the effect of NMP on biliary lesions not requiring intervention (ie, asymptomatic). Although not statistically significant, the lower MELD-Na score in the NMP cohort may reflect the tendency not to use NMP in more urgent cases in the setting of relatively new technology. However, there were no exclusion criteria based on MELD-Na, and we did, in fact, have 9 patients in the NMP cohort with MELD-Na ≥30 at the time of transplantation. In addition, our study did not seek to assess any specific indications for liver NMP, nor did it test the broad application of NMP in liver transplantation. Our study better reflects how NMP can be integrated into a mature North American transplant program.

Solid organ transplantation has been successful because of the ongoing commitment to innovation and development in all aspects of donor and recipient care, including surgical techniques, storage solutions, and immunosuppressive therapies. NMP provides another tool to facilitate liver transplantation, whether it is used as a substitute or in addition to traditional methods of SCS preservation. Our experience complements previous RCTs by showing that NMP can safely extend preservation without compromising graft or patient survival.

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Disclosure

The author of this manuscript has conflicts of interest to disclose as described by the American Journal of Transplantation. A.M.J. Shapiro reported serving as a paid consultant to ViaCyte Inc, Hemostemix Inc, and Aspect Biosystems Ltd. Other authors of this manuscript have no conflicts of interest to disclose as described by the American Journal of Transplantation.

Data availability

The data that support the findings of this study are available on request from the corresponding author.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ajt.2023.04.013.

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